

11. TMT System**TECHNICAL PARAMETERS**

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|---------------------------------|--|
| 1. Mill | Bar Mill |
| 2. Input Size | Billets 100 mm |
| 3. Sizes to be processed | 8 - 32 mm. |
| 4. Temperature after last stand | Max. 1000°C for 8 - 12mm dia.
Max. 1050°C for 16 - 25mm dia. |
| 5. Temperature difference | Less than 50°C between head and tail ends
after last stand |
| 6. Chemical Composition | As per IS 1786 : 2008 |
| 7. Aim | Production of High strength rebars as per
IS 1786 : 2008
Grades Fe 415, Fe 415D, Fe 500, Fe 550D,
Fe 550, Fe 550D and Fe 600. |

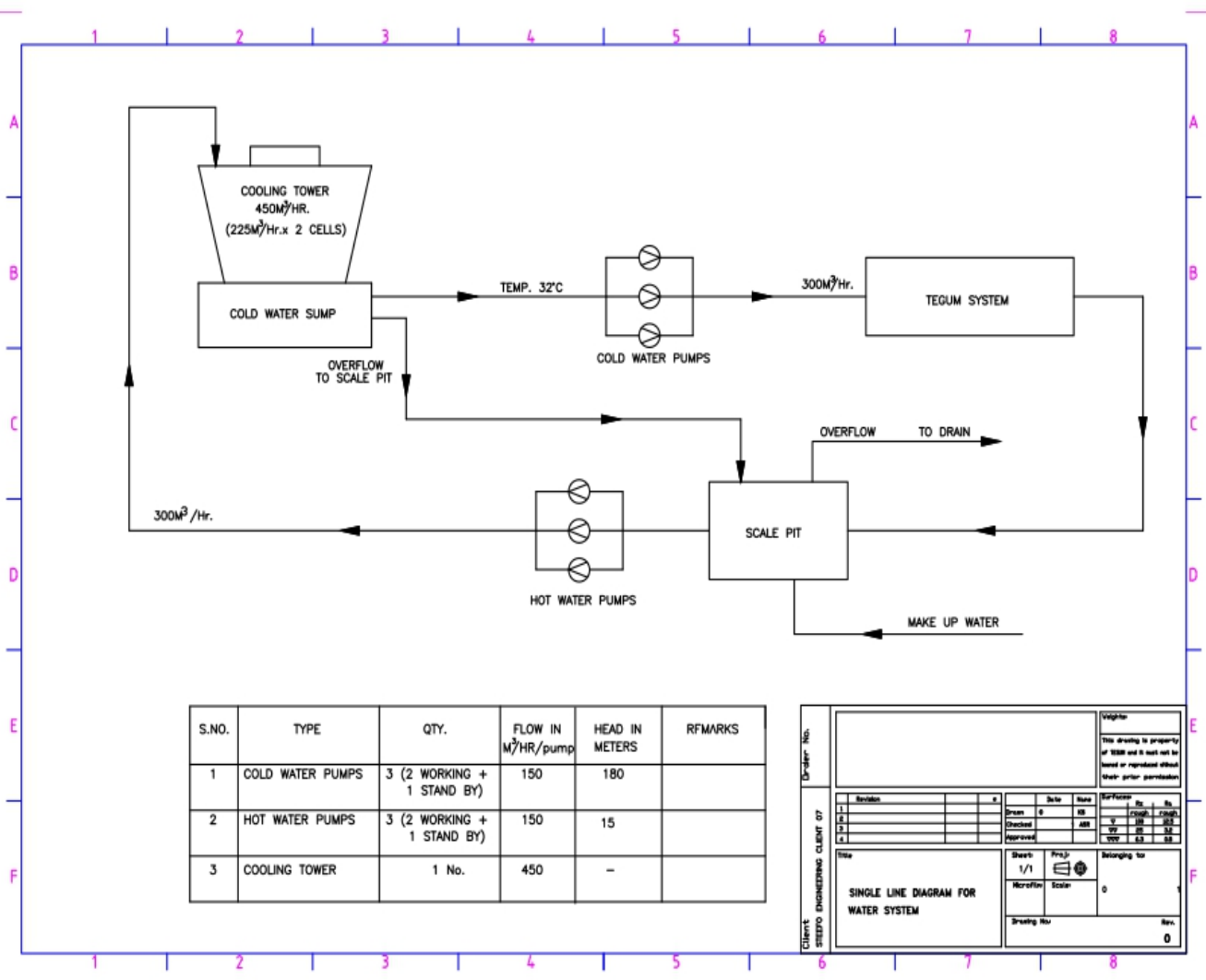


The diagram illustrates the TMT (Thermal Management System) components and their interconnections. Key elements include:

- Pressure Lines:** 100 N.B. AIR PRESSURE LINE at the top.
- Flow Indicators:** Arrows showing flow direction, such as "To Pyrometer" and "COOLING TOWER BASIN".
- Valves and Pumps:** Various symbols for control valves, pumps, and isolation valves.
- Legend:** A table defining symbols for different types of lines, tanks, and equipment.

SYMBOL	DESCRIPTION
1-2	PROCESS LINE
3	PUMP
4	VALVE
5	ISOLATION VALVE
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99	ISOLATION VALVE
100	ISOLATION VALVE

Water Diagram



TECHNICAL DESCRIPTION

1 Introduction

For the purpose of controlled processing of deformed steel bars by means of high pressure water, Rolling Mill TMT System will be installed in the Rolling Mill with the aim to produce 'Quenched and Self-Tempered' (QST) steel rebars. A suitable shear needs to be located after the Rolling Mill TMT System to cut the rebars to suit the Cooling Bed length.

2. Technical parameters

The bars produced through the Rolling Mill TMT System as per this offer shall produce rebars as per the IS 1786: 2008 (Fourth Revision) amendment 3 of March 2017 Grades Fe 415, Fe 415D, Fe 500, Fe 500D, Fe 550, Fe 550D and Fe 600.

3. Layout

3.1 Main Rolling Mill TMT system with Pinch Roll Unit.

The main TMT system consists of the Rolling Mill TMT Pipes installed after the finishing rolling stand but before the cooling bed. Two pinch roll units need to be located before and after the Rolling Mill TMT pipes.

3.1.1 Arrangement of Rolling Mill TMT pipes

For dimension range of Ø 8 to 25 mm, adequate number and type of Rolling Mill TMT pipes of different lengths will be used. The length of the Rolling Mill TMT Pipes would suit mill parameters.

Fabricated base frame trolley(s) for mounting of the desired Rolling Mill TMT pipes will be provided for the system.

In the TMT Line, Pinch roll units with speed control are needed - to ensure that the tail end of rebar is thermo-processed in the same way as the front and middle portion of the rebar to obtain uniform properties over the entire length.

3.1.2 Water

High pressure water pumps will be provided so that water at desired pressure is available for rapid water quenching of rebars to obtaining desired properties.

3.1.3 Compressed air consumption

- Air Compressor

About 250 CFM at 7 bar will be needed to be provided by client.

3.1.4 Pressure water pump station

The pump station has high-pressure water pumps, including one standby pump, which will deliver water into a main pipe.

A motor operated valve controls water flow.



3.1.5 Details of the water and air-piping scheme

The Rolling Mill TMT pipes are fed from the pressure water manifold through flexible hoses that are controlled by manually operated hand valves. Each rolling mill TMT pipe has two water inlets.

The valves have only two positions "closed" and "open". This makes the operation very simple for the operators.

At the opposite end of the water inlets is arranged the air inlet. A manually operated valve controls the airflow.

3.1.6 Measuring and control instrumentation**3.1.6.1 Control instruments**

Located in a control cabin.

3.1.6.2 Pressure transmitter

A pressure transmitter with indicator and 4 - 20 mA signal mounted on water header near TMT system. The water pressure would be displayed on the digital meter located in the control cabin.

3.1.6.3 Temperatures

The final rolling temperature TF (measuring range 300 - 1300°C) is measured ahead of the TMT pipe.

The equalizing temperature TE (measuring range 300 - 1300°C) is measured on the cooling bed. For quality control purposes, the temperatures are indicated in the control cabin.

3.1.6.4 Check of tensile properties

At the beginning of a rolling campaign the tensile properties should be checked on samples as soon as possible and related to the equalizing temperature TE. This test is made by means of a tensile testing machine.

3.2 Cooling Tower & Scale Pit

The water used for water quenching as per the TMT Technology is collected in a scale pit via water flumes. It is then cooled at a Cooling Tower for re-use in the Rolling mill TMT system.

The hot water pumps and the Cooling Tower are included from our Scope of Supplies. Additionally, we shall provide the detail engineering for the water flumes and scale pit to enable the customer to build them as per local standards and practices.

3.3 Electrics & Controls

The electrics will operate from the supply voltage of 415V AC $\pm 10\%$, 50 Hz $\pm 5\%$.



SCOPE OF SUPPLY

The quantity and technical data given in this Chapter is indicative and subject to change. These will be finalized during detail engineering to ensure that the client is able to produce TMT rebars of international standard.

1 Main Rolling Mill TMT system

- 1 lot Rolling Mill TMT pipes with appropriate number of intermediate lengths.
- 1 lot manual valves PN 21 for water
- 1 lot flexible hoses PN 16 for water
- 1 lot manual valves R 1" PN 6 for compressed air
- 1 lot flexible hoses DN 20 PN 6 for compressed air
- 1 lot analogue pressure gauge display 0-25 Kg/cm² for water
- 1 lot motor controlled valve PN 21

2 Support Trolley for Rolling Mill TMT Pipes

The support base frame before cooling bed is to accommodate the different Rolling Mill TMT pipe lengths with holes bored to suit for mounting. When using short pipes, the remaining will be bridged with cast iron funnels

3 Piping

The interconnected pipes, with flanges and fittings will be supplied by us.

4 Temperature measuring instruments

- 1 No. Optical pyrometer, range 300 - 1300°C located immediately after the finishing stand;
- 1 No. Optical pyrometer, range 300 - 1300°C located in the centre of the cooling bed;

5 Pressure Transmitter

- 1 No. Pressure transmitter range 0 - 35 Kg/cm² located on the water header near TMT System.

6 Electrical equipment

The electrical equipment will be suitable to operate from 415 VAC $\pm$ 10%, 50 Hz $\pm$ 5% at 40°C ambient.

7 Controls

Controls would be located in the control room and pulpit and would be suitable to operate at an ambient temperature of 30°C.

8 High Pressure Pumps

Lot of high pressure pumps (including 1 standby) for pumping water from Cooling Tower basin to Rolling Mill TMT System will be supplied by us.

9 Cooling Tower

A Cooling Tower of adequate rating will be supplied by us.

10 Hot Water Pumps

Lot of hot water pumps (including 1 standby) for pumping water from the scale pit to the Cooling Tower will be supplied by us.

